

Java Assignment: ATM Machine Simulation

Problem Statement

Write a Java program that simulates a simple ATM machine.

When the program starts, ask the user to enter a 4-digit PIN.

Correct PIN: 1234

The user has 3 attempts. If all attempts are used, print 'Your account has been locked.' and terminate.

After Successful Login

Initial balance: **2500.75**;

Display this menu repeatedly until Exit is selected:

===== ATM =====

1. Check Balance
2. Deposit
3. Withdraw
4. Show Account Status
5. Exit

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Menu Options

1. Check Balance

- Display the current balance.

2. Deposit

- Ask for a deposit amount. It must be greater than 0; otherwise print 'Invalid amount.'
- Update the balance if valid.

3. Withdraw

- Ask for an amount. It must be greater than 0 and not exceed the balance.
- If it does, print 'Insufficient balance.' Otherwise subtract it.

4. Show Account Status

- VIP Customer (≥ 5000)
- Regular Customer (1000–4999.99)
- Low Balance (< 1000).

5. Exit

Print 'Thank you for using our ATM.' and use the break statement to terminate the loop.

Invalid Menu Choice

- If the user enters any option other than 1–5, display 'Invalid option.'

Additional Requirements

- Use meaningful variable names.
- Use appropriate data types.
- Use one while loop, one do-while loop, one switch statement, multiple if-else statements, and one break statement.
- The program should continue working correctly after multiple deposits and withdrawals.

Bonus Challenges

- Count successful deposit/withdrawal transactions and display the total when exiting.
- If the withdrawal amount is 0, print 'Transaction cancelled.'
- If the balance becomes exactly 0, print 'Warning: Your account is empty.'
- Display the updated balance after every successful deposit or withdrawal.